KOR STRUCTURAL · ENGINEERING TOOLS

Structural models built from the drawings

31168 YMCA Langara and 31138 2170 W 1st · prepared 7 August 2026

Andrea's ETABS models were lost with a failed drive. Rather than re-enter the building's geometry by hand, this reads the concrete-outline DXF plans drafting already exports and produces an ETABS model covering every storey. It does none of the engineering — it removes the typing.

60 storeys populated (31168)	**897** wall panels, true thicknesses	**2,418** columns, sized & oriented	**128** floor plates, median 9,800 ft^2

What to open

Each sits in its project's `02 Engineering\02 Lateral Design\01 ETABS Models\`. In ETABS: **File → Import → ETABS .e2k Text File**.

FILE	WHAT IT IS
`31168-FROM-DRAWINGS.e2k`	**The one that matters.** Tower B and the rest of the building — 60 storeys, 897 walls, 2,418 columns, 128 plates. Nothing of this existed before.
`31138-FROM-DRAWINGS.e2k`	A gap-fill. You have already rebuilt this building's walls — 23 to 28 panels on every storey — so this adds only the 96 members the drawings show and your model does not, mostly at podium and parkade. Nothing you have modelled is duplicated.
`…-report.txt`	Every sheet, the storeys it landed on, what it produced, and every location that needed judgement.
`_drawing-renders\`	The drawings themselves, with the extracted members drawn over them. Quickest way to see what was read and what was not.

Everything generated is named `KW*`, `KC*`, `KP*` with new sections prefixed `KOR-`, so you can select, filter or delete the whole lot in one action. Where the model already had a section of the right thickness it was reused, so members carry the project's own concrete and names you recognise.

What has been verified, and what has not

Verified: every member sits exactly where the drawing draws it

Each plan is rendered with the extracted members drawn over it (`takeoff dxf-render <plan> <png> --overlay`), so what the model carries can be compared against the linework directly rather than inferred. On B-LEVEL 30 the core's two shafts, its bar wall and all twenty-four perimeter columns land on their outlines; the images are in `_drawing-renders\`. Wall thicknesses and column sizes are measured from the outlines, not assumed.

Verified: against your own model

On 31138 the generated columns land where you placed yours by hand: no shift applied, 43 of your 87 columns matched within six inches, and on L10 the median distance was 0.0. On 31168 the wall inventory agrees with the model that was lost — its pier-force export records five wall piers on every typical storey and 34 at L01, and the drawings give five core elements per typical storey and 29 walls at L01.

Verified: the geometry lands on your grid

The drawings and the model both come out of the same Revit project, so they already share a coordinate system and the geometry is placed without any shift. That is checked against the grid itself rather than assumed: the tower core straddles grids 15 and 16, which ETABS places at x = 1500 and 1826, and the walls read out of the drawing sit at 1500.05 and 1826.05 — a twentieth of an inch. The same holds up the tower, with wall coordinates landing on grid lines on B-LEVEL 30, B-LEVEL 33 and LEVEL 20.

Worth knowing when you look: only the inner face of a core wall sits on its grid line, the outer face standing off by the wall thickness, which is how the drawing draws it.

What it reads, and how

LAYER	BECOMES	HOW
`JBP_V-WALL`	Wall panels	A core is drawn as one ribbon tracing both faces of a group of walls. Each face is paired with the face opposite it; the pair gives the centreline and the true thickness.
`JBP_V_COL`	Columns	Footprint gives size and rotation, so a 16×24 sits the way it is drawn.
`JBP_C_SLABEDG*`	Floor plates	Outlines stitched into rings; rings inside a ring are openings.

Sheet titles carry the placement. `LEVEL 29 PLAN (L29-35)` fills seven storeys; `A-LEVEL 28` goes to Tower A alone; `LEVEL P2` to the parkade. The geometry is merged into an `.e2k` that ETABS itself exported, so storeys, grids, materials and design settings remain ETABS's own — the file differs from a known-good one only by the lines added.

Limitations — read before you rely on it

Slab plates are the weak half

Drafting interrupts slab edges wherever other linework crosses them, so many outlines do not close. Those are dropped and reported rather than guessed at, and rings under 50 ft² are discarded as noise — the 128 plates that survive have a median area of 9,800 ft², which is a real floor plate rather than a sliver. Coverage is therefore partial: the plates that are there are believable, but not every storey has one. Treat them as a starting point, or ignore them and place your own diaphragms.

Widening the closure tolerance was tried and rejected on the evidence: at 18″ the model gained 302 more plates, but the count of plates in a plausible size range did not change at all — every addition was a fragment. Recovering the missing plates needs a better method, not a looser one (see below), and that is why only one model is shipped.

What is set up, and what is yours to decide

The model arrives load-ready rather than load-less. Load patterns (Dead with self-weight, Live), load cases (Modal at 55 modes, Dead, Live) and the mass source come through from the reference, and because every wall, column and plate now carries its true thickness and the project's own concrete, **self-weight is real the moment you run it**. On 31138 your own 149 area-load assignments are preserved untouched.

What is not applied is superimposed dead and live load on the generated plates, and that is deliberate. Your own 31138 model carries five different SDL values on L01 alone — 5, 25, 45, 70 and 145 psf — with live load from −50 to 150, because they follow occupancy: terraces, planters, amenity, suites, corridors. None of that is visible in a structural outline, and putting a storey's "typical" value on a landscaped terrace would flow straight through to the foundation loads.

A rigid diaphragm per storey is assigned to every generated plate, so modal analysis runs meaningfully out of the box. Also yours: spectra, stiffness modifiers, pier and spandrel labels, and meshing.

Openings are detected but not cut

Shafts and stair openings are found as inner rings and reported, but not subtracted from the plates, so slab areas read high where a core passes through.

> **Check the thick walls.** Anything modelled thicker than 24″ is flagged in the report. Wall outlines that meet at junctions can pair across the junction and read thicker than the wall really is. The flags tell you exactly where to look.

31168's storeys are the whole site, not Tower B alone

The model that was lost was Tower B on its own, storeys L01 to L40 with Mezz and P1–P2. This one is built on the site model exported from Revit, so its storeys are the whole development — `B-LEVEL 27` upward for the tower, shared unprefixed storeys lower down. The geometry is right; the frame it sits in is not the one you were working in. Worth deciding before you build on it.

It believes the drawing

Thicknesses and sizes come from what was drawn, not from a schedule. A wall drawn at the wrong width models at the wrong width. The plans used here are dated 28 June 2026 — later revisions are not in this model.

The questions, in one place

Beside each model is `…-QUESTIONS-for-Andrea.xlsx`. It lists every decision the tool had to make without being told — wall or pier, doorway or end of concrete, which storeys need plates, what superimposed load belongs where — with what it did, why it matters, and the evidence behind it. There is an answer column.

Those answers are the point. Each one becomes a rule the tool applies from then on, on this project and every one after, instead of a judgement folded into the code where nobody can see it. The workbook also carries every flagged location and a ledger of which drawing filled which storey, so a wrong answer is traceable to the sheet that caused it.

What would make it materially better — your input decides

IMPROVEMENT	WHAT IT NEEDS FROM YOU
Cut the openings. Subtract shafts and stairs from the plates so slab areas and diaphragms are right.	Confirm that inner rings on the slab-edge layers are always openings, never a step or a depression.
Close slab edges properly by extending an interrupted edge along its own direction to the line that cut it, instead of bridging by distance. Measurement says this is the only way to recover the missing plates.	Nothing — but tell us which storeys you most need plates on, so it is proven where it matters.
Wall grades by height. Read the wall schedule so 65 MPa low and 45 MPa high come through instead of one material.	Point us at the schedule sheet you would read for this, and how you would expect the grades keyed to walls.
Beams. Nothing on the beam layers is read today.	Whether beams are worth having, or you would rather add them yourself.
A button in the KOR app instead of a command line.	Whether you would run this yourself per project, or ask for it.

The single most useful thing you can send back: open the model, and tell us the first three things that are wrong or annoying. Each one is a rule the tool does not yet know.

Where this sits in the KOR toolset

This is a new front end on the engineering engine the app already carries, not a separate program.

```
Revit  ──▸  drafting's DXF plan exports  ──▸  [ DXF → ETABS ]  ──▸  .e2k  ──▸  ETABS
PDF drawings ──────────────────────────────▸ [ PdfToSafe   ]  ──▸  .f2k / .e2k / live OAPI
PDF drawings ──────────────────────────────▸ [ Takeoff     ]  ──▸  quantities, xlsx
```

All three live in `Kor.Operations.EngineeringTools`, share the same geometry code, and are covered by the same test suite (384 tests). PdfToSafe reads drawings when only a PDF exists; this reads the DXFs when drafting has exported them, which is exact and needs no vision model. The natural next step is a button in the app's Engineering Tools panel so it runs without a command line.

> **It only stays reliable while the layer names hold.** `JBP_V-WALL`, `JBP_V_COL` and `JBP_C_SLABEDG` are now a machine contract, not just a drafting habit — which is a concrete argument for the Revit template and standards work already under way.

Built 6 August 2026 · `Kor.Operations.EngineeringTools.Core/Dxf` · CLI `takeoff dxf-to-etabs`, `dxf-inspect`, `e2k-compare` · documented at `docs/DxfToEtabs.md`